

## Lecture Four

CSCI Technology Innovation | W26  
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## The Problem Space

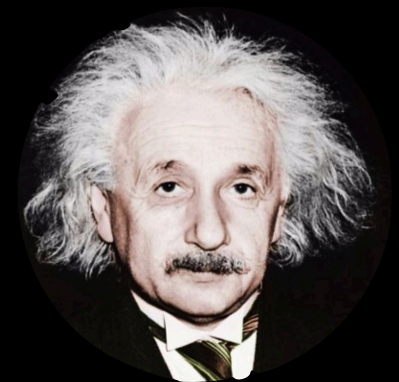
The Single Hardest Part of Innovation

**Learning to live in the problem space**

**“Falling in love with the problem not the solution”**

*“If I had an hour to solve a  
problem, I'd spend 55 minutes  
thinking about the problem and  
five minutes thinking about  
solutions”*

- Albert Einstein



The Single Hardest Part of Innovation

Learning to live in the problem space

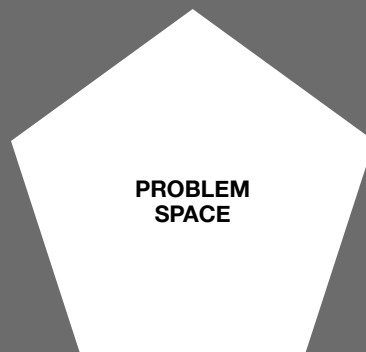
“Falling in love with the problem not the solution”

**Staying in the problem space  
throughout the life of the product**

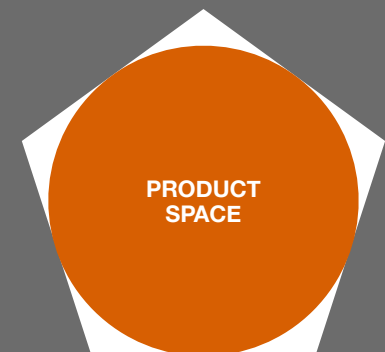
| COMPANY | PROBLEM SPACE                              | SERIES OF SOLUTIONS                                               |
|---------|--------------------------------------------|-------------------------------------------------------------------|
| AMAZON  | One-stop shopping that's convenient & fast | Web -> closer warehousing -> Prime -> returns -> drones           |
| SPOTIFY | Discover something to listen to            | Search -> playlists -> algorithmic discovery -> podcasts -> AI DJ |
| NETFLIX | Home delivery of premium entertainment     | DVDs by mail -> streaming -> original production -> games         |

**Staying in the problem space  
throughout the life of the product**

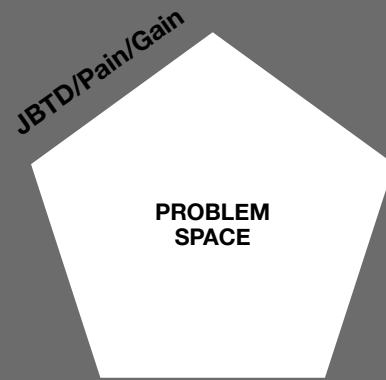
Giving shape to the product



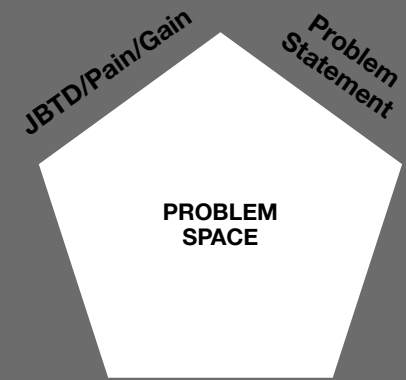
Giving shape to the product



Giving shape to the product



Giving shape to the product



*"I'm always behind on my coursework. I try to keep up with the lectures, but I have three different classes with assignments due at different times plus three shifts a week for my part-time job. I call home at least once a week, I run every day, and I have to find time to eat. I often pull all-nighters to finish papers or cram for tests because I need to get good grades.*

*Planners are useless. I keep forgetting to update it. Funny that a something meant to be a time-saver is a waste of time.*

*By the time midterms come, I'm exhausted and stressed. My grades are suffering and that makes me more stressed."*

JBTD/Pain/Gain

From desire to why to what

Problem Statement

Customer Discovery

A time-starved university student needs a way to feel in control of their academic and personal commitments because they want to avoid missing critical activities.

*<A user> needs a way to <desire or goal> because they want to <insight about why it matters>*

### Problem Statement

*NOT*

Students need a smart calendar because they want something easier than existing planners.

### THE BIG IDEA

The problem space persists throughout the life of the product and begins with a problem statement from the point of view of a user.

### Transforming from a group to a team

Talking about other than tasks

Having regular check-ins

Knowing each other's strengths/preferences

Knowing areas where each is less comfortable

Having a common goal

**THE BIG IDEA**

A group transforms into a high-performing team when members feel they can trust each other (“psychological safety”).